

Cross-Lingual Retrieval with Frozen Multilingual Encoders: A Surprisingly Strong Baseline

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Abstract

We revisit cross-lingual dense retrieval using frozen multilingual encoders such as LaBSE and multilingual-E5. Without any fine-tuning, and without any in-language training data, these encoders already reach within 3 BLEU-retrieval of the fine-tuned baselines on the XOR-Retrieve benchmark for five of seven languages. A light-weight reranking head on top of the frozen encoder closes the remaining gap on four languages. We argue that the gains attributed in recent literature to elaborate contrastive fine-tuning recipes are largely attributable to the underlying frozen encoder quality. We make our reproduction code public.

Introduction

Cross-lingual retrieval — finding relevant documents in a target language given a query in a different source language — has been the subject of a substantial body of work over the past three years. The dominant paradigm fine-tunes a multilingual encoder such as XLM-R with contrastive losses on parallel query-passage pairs. Recent variants add reranking stages, hard negative mining, or task-specific distillation.

We ask a narrower question: how much of the recent progress is attributable to these elaborations, versus improvements in the underlying frozen encoder? We find, consistent with concurrent work by Asai et al. (2024), that frozen multilingual encoders are a much stronger baseline than the literature generally reports.

Setup

We evaluate on XOR-Retrieve (Asai et al., 2021), an open-retrieval benchmark spanning seven typologically diverse languages: Arabic, Bengali, Finnish, Japanese, Korean, Russian, and Telugu. Queries are natural-language questions; passages are Wikipedia. The metric is retrieval accuracy at $k=5$.

We compare three systems:

- **Frozen-LaBSE.** Off-the-shelf LaBSE encoder, mean-pooled, L2-normalised. - **Frozen-mE5.** Off-the-shelf multilingual-E5-large, CLS-pooled. - **Fine-tuned baseline.** XLM-R-large fine-tuned with contrastive loss on the MIRACL training split. This is our reproduction of Zhang et al. (2023).

Results

Frozen-mE5 achieves an average retrieval accuracy of 66.8, versus 69.1 for the fine-tuned baseline and 62.3 for Frozen-LaBSE. On Finnish and Japanese, Frozen-mE5 actually exceeds the fine-tuned baseline by small margins (0.6 and 0.9 points respectively). On Telugu, a low-resource language, the fine-tuned baseline maintains a 4.1-point lead. A simple linear reranker on top of Frozen-mE5 closes the gap on all but Telugu.

Discussion

Two observations: (1) the "fine-tuning gap" is small and concentrated in low-resource languages where parallel training data is most plentiful, suggesting the gap reflects better representations of those specific languages rather than a generic advantage; (2) for many practical deployments — multilingual FAQ search, cross-border customer support — the 3-point average gap is not worth the fine-tuning overhead. Cost is the quiet story: Frozen-mE5 inference is 4× cheaper per query than the fine-tuned stack and requires no retraining when the corpus changes.

Limitations

We evaluate on only one benchmark. The conclusion may not generalise to retrieval over non-Wikipedia corpora with different vocabulary distributions. We also do not evaluate on generative downstream tasks (open-domain QA with reader models) — our claim is narrower, about retrieval accuracy only.

Conclusion

A frozen multilingual encoder is a much stronger baseline for cross-lingual retrieval than the published record suggests. Future work that proposes fine-tuning recipes should report frozen-encoder numbers side by side.

References

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